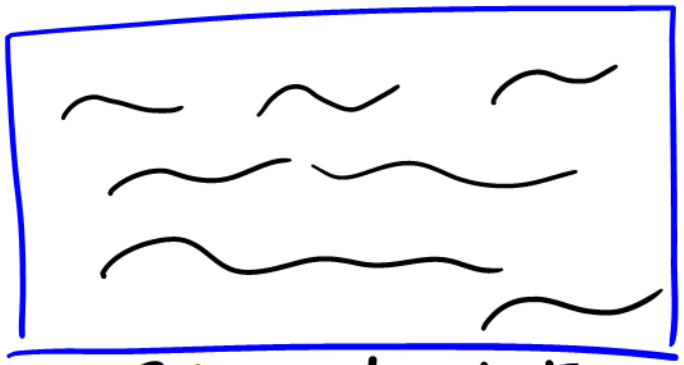
Tendon characteristics

Tendon transmits force of associated muscle to bone

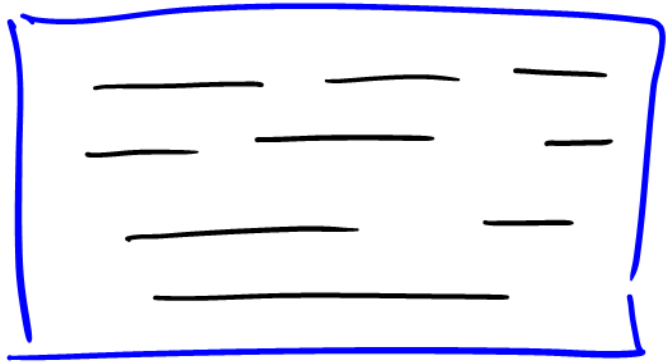
Tendons are viscoelastic, non-linear

flexible

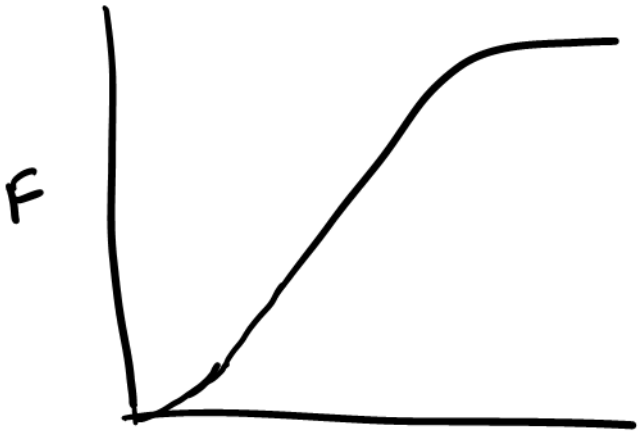
Cord or strips - circular, oval or flat



Relaxed state

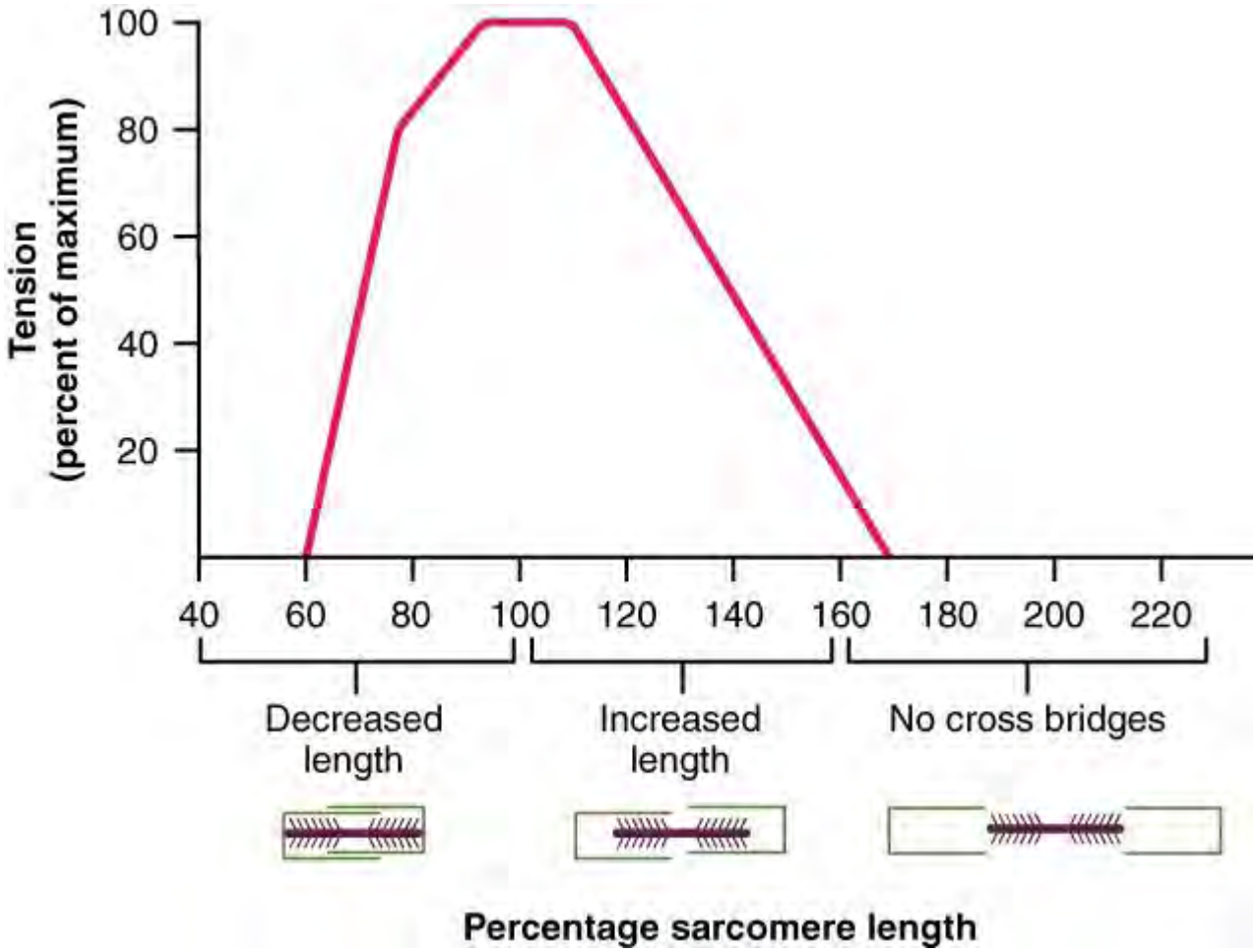


Stretched state



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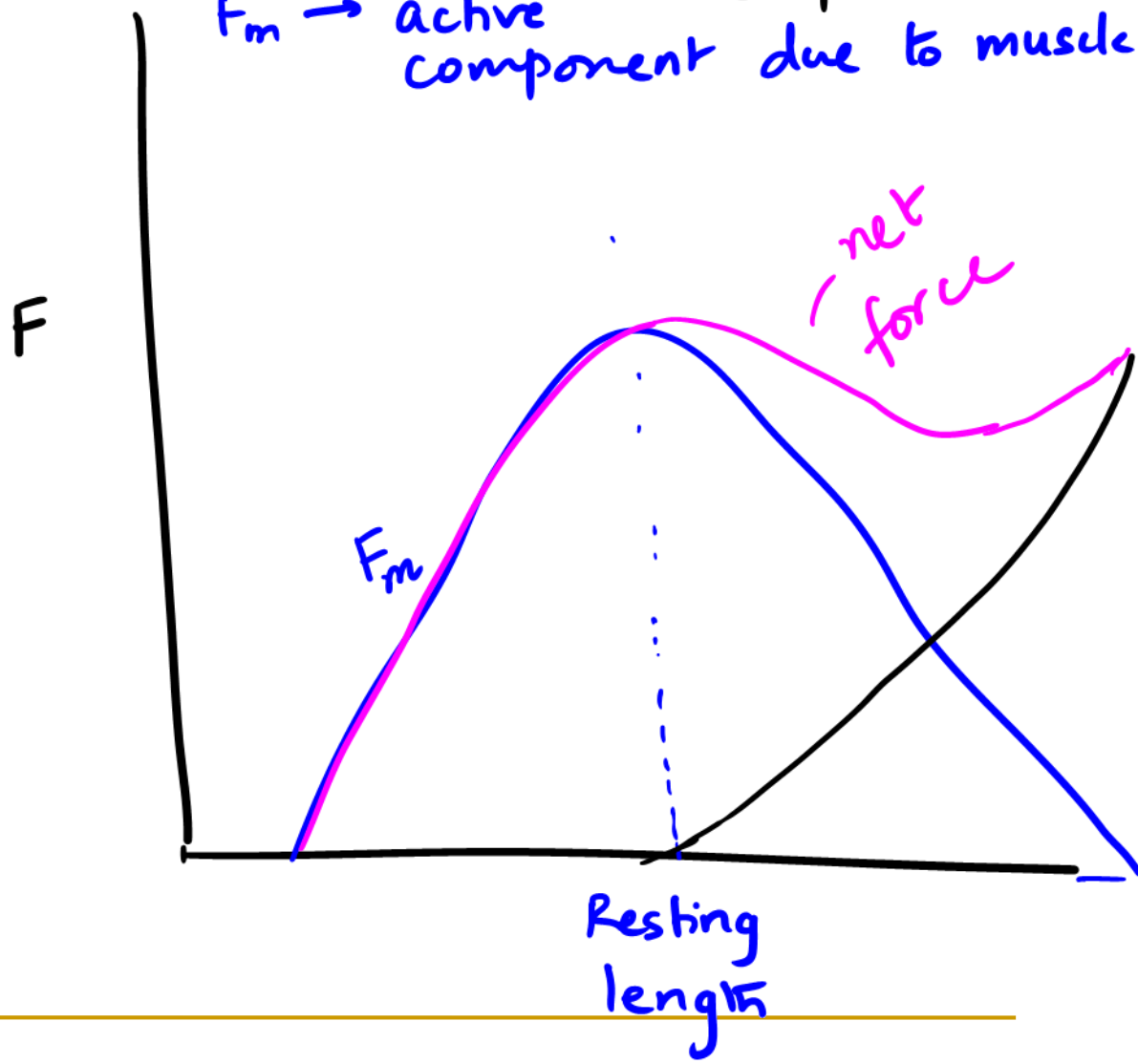
Length-Tension relationship



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Connective tissue
Tendons
] Passive Elastic components

F_m → active component due to muscle



Biarticular muscles cross two joints

Passive tension starts playing a role

Active insufficiency – two-joint muscles cannot contract maximally across both joints at the same time

Finger & wrist flexors

Hamstrings – hip extension limited when knee is flexed

Passive insufficiency

Occurs with biarticular muscles

They cannot fully stretch over both joints at the same time

Biarticular muscles

Hamstrings

Rectus femoris

Gastrocnemius

Flexors & extensors of the hand

ORIGIN

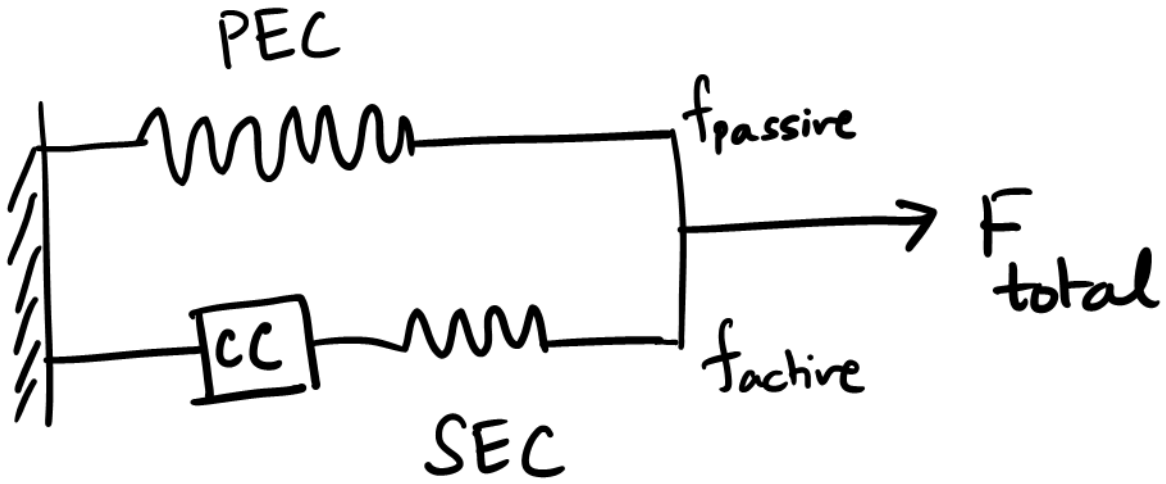
More proximal attachment or closer to the middle of the body

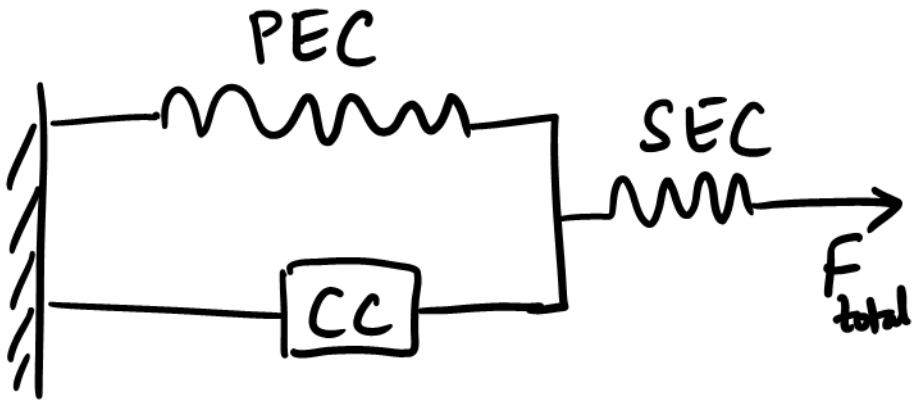
INSERTION

Attachment that is distal or farther from the midline

Hill's muscle model

Musculotendinous unit





CC → contractile component

SEC → series elastic component

PEC → parallel elastic component

SEC usually stiffer than PEC